

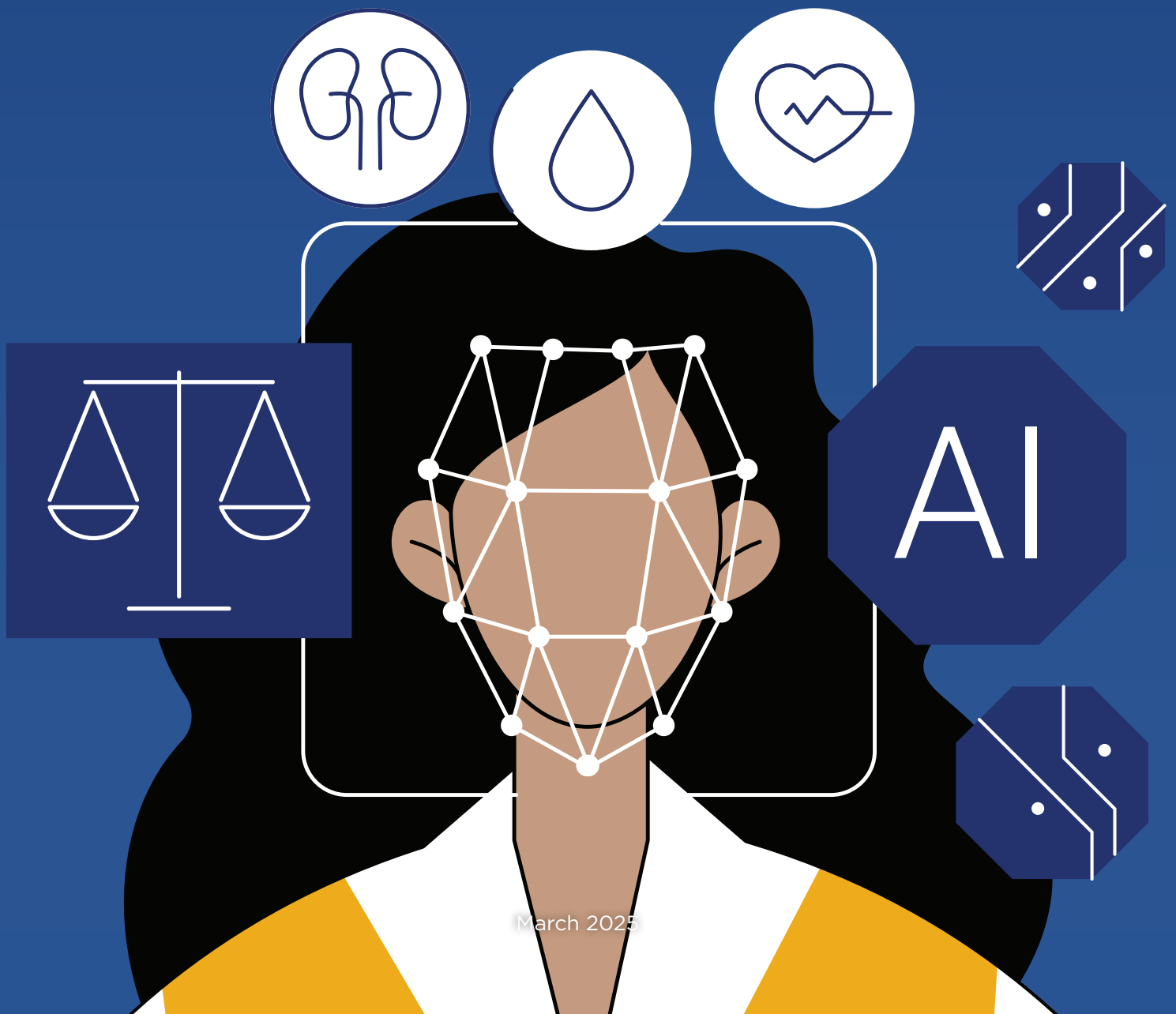


MENA OBSERVATORY
ON RESPONSIBLE AI
مركز الشرق الأوسط وشمال أفريقيا للأبحاث الاصطناعية المسؤول

Governing Responsible Artificial Intelligence and Data
in the Middle East and North Africa (MENA)



AI and Women's Health Data Governance Case Study of Morocco



March 2025



AI AND WOMEN'S HEALTH DATA GOVERNANCE CASE STUDY OF MOROCCO

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INTRODUCTION

Data governance in the era of artificial intelligence (AI) has been central to Morocco's digital transformation strategy since the current government's appointment in 2021 and the establishment of the Digital Transformation and Public Administration Ministry. This case study focuses on the use of AI tools in the governance of health data, specifically examining women's data in the Marrakech-Safi region. The study is based on a survey consisting of 40 questions administered to 10 organizations from both the public and private sectors. The case study provides an evaluation of current practices related to data collection, processing, sharing, storage, and security.

The purpose of this case study is to provide empirical evidence on existing health data governance practices and frameworks, identify challenges and gaps, and explore opportunities. Additionally, it aims to produce policy-oriented recommendations to enhance responsible data governance for AI in Morocco's health sector.

In this case study, we considered the diversity of actors and types of data collected and the methods used for data collection, storage and security, including AI and digital tools as well as traditional methods. We adopted a multisectoral approach in selecting survey participants to ensure our data will effectively represent the public and private sectors, associations, and patients.

Our understanding of data on health in this study takes its diversity in form into consideration. Health data can be in the form of health records or clinical data, demographic data, personal data, anonymous data, or insurance claims.

METHODOLOGY

To gather data, a survey was conducted using semi-structured questions. The questionnaire was tailored to each organization and participant based on their role (e.g., nurses, doctors, data governance officers, authorities in public and private hospitals and patients). Participants were also invited to provide information on any parameters not covered by the questionnaire. The survey included open-ended questions to capture institutions' experiences with AI and data governance.

The study's scope, like any empirical questionnaire-based research, was inherently limited by the number of participants and types of questions posed. In this case, the survey targeted 10 organizations encompassing public and private sectors, associations, and individuals, with re-

spondents required to answer 40 questions. A multi-sectorial and gender-sensitive approach was employed in both the design of the questionnaire and the selection of participants to guarantee diverse perspectives.

The collected data was analyzed using both quantitative and qualitative methods, and the results are presented in this case study report, which highlights key findings and provides recommendations for policymakers and stakeholders.

The report is structured around the World Bank's theoretical framework, which identifies five core data governance practices: data collection, processing, sharing, storage, and security. These practices are analyzed with reference to three primary objectives: protecting individuals, promoting health value, and prioritizing equity.

BACKGROUND

The COVID-19 pandemic accelerated the adoption of AI across various sectors, including food security, health services and sustainable development. AI systems are used on different levels, from automation to intelligent drug design and AI-powered surgical robots. The largest portion of research around AI in healthcare is dedicated to its use in clinical trials and robot-assisted surgery, followed by connected machines and fraud detection.

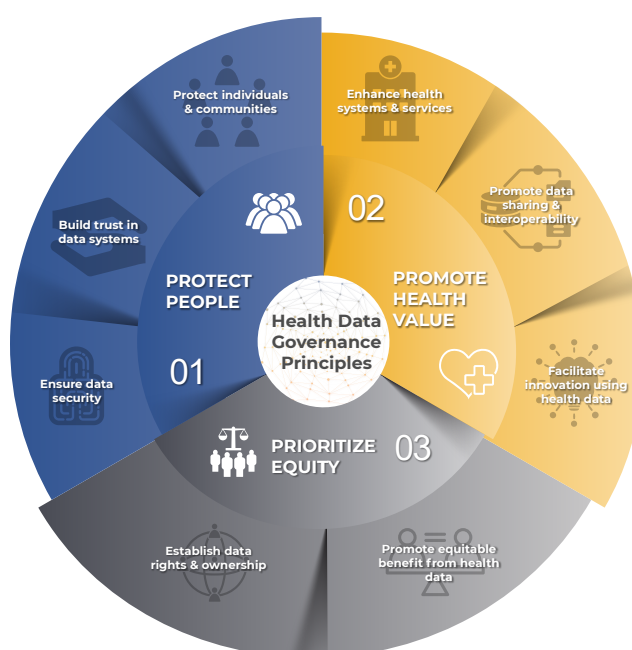


Figure 1: The Health Data Governance Principles framework.¹

¹ The World Bank's Health Data Governance Principles (<https://healthdatagovernance.org/principles/>), revisualised

In light of the benefits and challenges brought by AI and digital transformation, Morocco established the Agency for Digital Development in 2017. The agency's mandate is to implement the state's strategy for digital development and promote digital transformation across all sectors, including health-care.² In his 2020 Throne Day speech, King Mohammed VI announced a plan to expand health insurance coverage in Morocco from 25 percent to 100 percent of the population by 2025. The initiative was significant, given that 44.6 percent of women in Morocco currently lack health insurance coverage.³

Analyzing health data governance in the age of AI begins with a review of conceptual and theoretical frameworks that help understand both international and national contexts. This analysis focuses on two key tensions: first, between local and global perspectives, and second, between urban and rural organizations and the public and private sectors. These tensions influence health data governance and steer global and national policy actions in response to new challenges. AI and digital transformation bring both opportunities and challenges, particularly in the context of health data governance. These technologies have the potential to reshape health data governance, particularly in terms of their implications on health actors, regulations, and policy actions.⁴ But they also raise critical concerns regarding the roles and responsibilities of health actors, the adequacy of existing regulations, and the development of policies that effectively address issues such as data privacy, security, and equity. Understanding these dynamics can guide us toward a governance framework that balances national rules and legislations with international

law, especially human rights law, international health law, and ethical standards for AI.⁵

Health data governance in the era of AI has created tensions over data sovereignty among states. Such challenges underscore the need for multilateral and bilateral cooperation, particularly in the wake of COVID-19, which accelerated debates over health data governance amid the growing influence of AI. Disputes over data sovereignty have emerged both between countries, such as the USA and China, and within nations, among health organizations operating at various levels of the health-care system.⁶ As states race to develop, regulate, and implement national AI strategies, it is crucial that health data governance remains a central focus of any national AI strategy, ensuring data privacy, security, and ethical use in healthcare innovation.

Morocco has prioritized women's health through efforts to improve healthcare access and address various challenges, particularly by leveraging digital transformation in health data governance. Digitizing traditional data processing in the sector has significantly impacted various aspects of women's health, including maternal health, family planning, reproductive health, and cancer awareness.

The government has launched initiatives to reduce maternal mortality rates and improve access to prenatal and postnatal care, working to enhance healthcare infrastructure and services in rural areas, where access is more limited. Family planning services are available, supported by programs to increase awareness and access to contraceptives. However, cultural and religious factors continue to influence family planning decisions.

Efforts to address reproductive health issues in Morocco form part of a broader healthcare framework, emphasizing access to information and services related to reproductive health. Key initiatives include raising awareness about sexually transmitted infections. Collaborations with

The government has worked to enhance healthcare infrastructure and services in rural areas, where access can be more limited to ensure women's rights to healthcare.

2 Agence de Développement Digital, *Smart Government* (2023), <https://www.add.gov.ma/smart-gouvernement>

3 Kawtar Zahidi et al., "Universal Health Coverage in Morocco: The Way to Reduce Inequalities: A Cross-Sectional Study," *Open Public Health Journal* 15 (2022), <https://openpublichealthjournal.com/contents/volumes/V15/e187494452212220/e187494452212220.pdf>

4 Tom Christensen and Per Lægrend, "The Whole-of-Government Approach to Public Sector Reform," *Public Administration Review* 67 (2007): 1059–66, <https://onlinelibrary.wiley.com/doi/10.1111/j.1540-6210.2007.00797.x>

5 Khoulood Bennani, Anissa Ben Hassine, Bachir Mazouz, "The Tensions of Public Governance: A Managerial Artifacts-based View," *International Review of Administrative Sciences* (2021), <https://doi.org/10.1177/0020852320988155>

6 Ibid.

other actors, women's associations and NGOs in particular, focused on improving health education on a variety of issues, including breast cancer awareness. Campaigns were held to underscore the importance of regular check-ups, nutrition, and preventive practices to encourage early detection and treatment. Associations and government agencies have also worked to reduce the stigma associated with breast cancer.

RESULTS

I. AI AND HEALTH DATA GOVERNANCE: LEGAL FRAMEWORK AND POLICY ACTIONS

AI's role in data governance has grown significantly, particularly since COVID-19, further complicating an already complex field, especially regarding the very definition of "data." The scope of health data, encompassing both personal and clinical information,⁷ exemplifies this complexity, as a primary challenge arises from the absence of a universally accepted definition to data.⁸

The World Bank Group highlights the multifaceted nature of data: it "may be quantitative or qualitative; stored on analog or digital media; collected through observation, as part of digital transactions or simply as by-products of digital lives; processed, structured, and analyzed to be converted into information (data are not synonymous with information); about people, things, and systems."⁹

The United Nations Secretary-General's Roadmap for Digital Cooperation emphasizes the critical need to protect personal data in the health sector. It highlighted how the digitization and secure exchange of health data among specialists can enhance patient care by providing an accurate understanding of the patient's health status, aiding doctors at providing more precise treatments.

In the wake of COVID-19, the World Health Organization stated that member states "are obliged under the International Health Regulations to develop public health surveillance systems that capture critical data for their COVID-19 response while ensuring that such systems are transparent, responsive to the concerns of communities,

and do not impose unnecessary burdens, for example, infringements on privacy."⁹

Artificial intelligence systems are increasingly being utilized to address challenges posed by the growing number of patients with multimorbidity.¹⁰ They are also utilized across various healthcare levels, ranging from automation processes to advanced applications like intelligent drug design and AI-powered surgical robots.¹¹ The largest share of the global AI market in healthcare is focused on clinical trials and robot-assisted surgeries, followed by connected medical devices and fraud detection systems.¹² These advancements are being further accelerated by technologies such as 5G and telemedicine.

II. LEGAL FRAMEWORK ON DATA GOVERNANCE IN HEALTH SECTOR

Morocco's 2011 constitution recognizes the right of all citizens to healthcare services, the right to respect privacy, and the right to protect personal data, including data pertaining to health.

The Moroccan legislator established basic principles for the protection of personal data through a set of laws and decrees, before issuing a more comprehensive law, Law no. 09.08, on the protection of private persons and their personal data.¹³

Decree 1026-97-2 of 27, issued on February 25, 1998, defines general regulations for public telecommunications networks,¹⁴ prioritizing confidentiality and neutrality in transmitting messages. According to the first paragraph of Article 5, the decree establishes provisions for safeguarding beneficiaries' private lives and nominal data.¹⁵ These regulations can only be overridden under specific legal conditions. Furthermore, telecommunications operators are obligated to inform their employees about these confidentiality requirements and the legal repercussions of failing

7 World Bank, *Implementation Knowhow Brief* (2021), 9, <https://openknowledge.worldbank.org/server/api/core/bitstreams/2dad9719-f398-4414-ae58-e4f3eca16f26/content>

8 Ibid.

9 Ibid.

10 Indrajit Hazarika, "Artificial Intelligence: Opportunities and Implications for the Health Workforce," (2020): 241, <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC7322190/>

11 Indrajit Hazarika, "Artificial Intelligence: Opportunities and Implications for the Health Workforce," (2020): 241, <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC7322190/>

12 Grand View Research, *Artificial Intelligence In Healthcare Market Size, Share, and Trends Analysis Report By Component (Software Solutions, Hardware, Services), by Application (Virtual Assistants, Connected Machines), by Region, and Segment Forecasts, 2022-2030* (2022), www.grandviewresearch.com/industry-analysis/artificial-intelligence-ai-healthcare-market

13 Law 08-09 on the Protection of Private Persons vis-à-vis the Processing of Personal Data, Dahir No. 1.09.15 of 22 Safar 1430 (February 18, 2009), <https://www.cndp.ma/images/lois/Loi-09-08-Ar.pdf>

14 Decree 1026-97-2 of 27 Shawwal 1418 (February 25, 1998), relating to the General Conditions for the Exploitation of Public Telecommunications Networks, as amended by Decree No. 2-05-771 of 6 Jumada II 1426 (July 13, 2005), and Decree No. 2-06-418 of Ramadan 17, 1427 (October 10, 2006)

15 Ibid.

C



to uphold the confidentiality of correspondence, including matters related to health data.

Law 08-09, issued on February 18, 2009, establishes protections for individuals regarding the processing of their personal data,¹⁶ aiming to prevent excessive and irresponsible use by both private and public institutions. This includes safeguarding sensitive information such as telephone numbers, national ID numbers, or genetic data.

The law covers the following key areas:

- Rights of the data subject
- Obligations of data controllers
- Role of the National Control Commission for the Protection of Personal Data (CNDP)
- Transfer of data to foreign countries
- Penalties

However, the scope of this law's application remains restricted, as the Moroccan legislator has excluded specific types of data processing, including the processing of personal data by private individuals strictly for personal or do-

mestic activities, data related to national defense and state security interests, processing aimed at the prevention and suppression of crimes, and data obtained and processed under specific legislative provisions.

The adoption of the 2011 Constitution marked a significant milestone in personal data protection in Morocco. By enshrining the right to privacy as a constitutional principle, the legislator elevated the protection of individuals' personal lives to the highest legal standard. Article 24 explicitly states, "Everyone has the right to protect his private life."¹⁷

This article highlights the prominence accorded to the protection of private life within Morocco's constitutional framework. This protection is further supported by the preamble to the 2011 Constitution, which affirms the supremacy of international conventions over national legislation.

Additionally, Article 27 of the 2011 Constitution states that "the right to information may only be restricted by law, with the aim of protecting everything related to national defense, protecting the internal and external security of the State, the private life of individuals, as well as preventing infringement of the fundamental freedoms and rights enshrined in this Constitution."¹⁸

In addition to the legal provisions contained in the Code of Criminal Procedure, particularly Article 108, which prohibits capturing, copying, or seizing of calls and communications made via remote communication, chapter ten of part one of Book A of the Criminal Code establishes further requirements related to the protection of individuals' personal data. Relevant legal texts include:

- Dahir No. 1.11.03 of 14 Rabi' al-Awwal 1432 (February 18, 2011) implementing Law 31-08, which establishes measures for consumer protection¹⁹ and includes provisions for privacy protection added to the Penal Code.

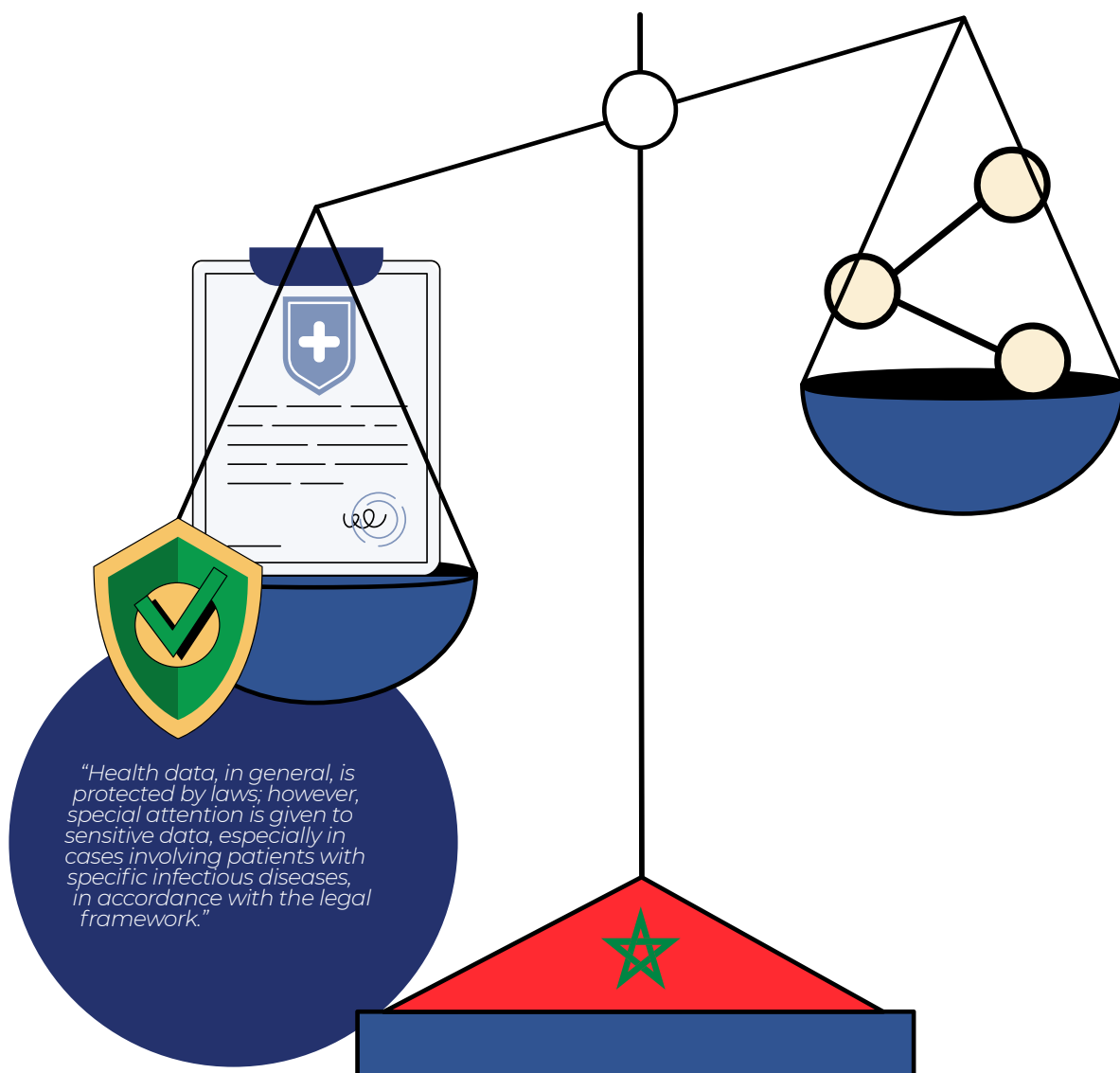
Other laws that address data protection and privacy include:

¹⁷ Secrétariat Général de Marocaine Gouvernement, *La Constitution Marocaine de 2011* (2011), https://www.cndp.ma/images/lois/constitution_2011_Ar.pdf

¹⁸ Ibid.

¹⁹ Dahir No. 1.11.03 of 14 Rabi' al-Awwal 1432 (February 18, 2011), implementing Law No. 31-08 Establishing Measures for Consumer Protection, accessed August 21, 2024, https://www.cndp.ma/images/lois/Loi_31-08_Protection-du-Consommateur_Ar.pdf

¹⁶ Law 08-09 on the Protection of Private Persons vis-à-vis the Processing of Personal Data, Dahir No. 1.09.15 of 22 Safar 1430 (February 18, 2009), <https://www.cndp.ma/images/lois/Loi-09-08-Ar.pdf>



- Law 53-05 on the electronic exchange of legal data²⁰
- Law 24-96 on Posts and Telecommunications²¹
- Law 03.07 on the disruption of automated data processing systems²²
- Press Law: Articles 38 and 44 to 51 bis of the Press Code, which criminalize violations such as insults, slander, or harm to individuals' private lives through the media²³

- Rules of procedure of the National Control Commission for the Protection of Personal Data

The aforementioned articles criminalize offences such as insulting, defaming, or infringing upon individuals' private lives through the media. By enacting various provisions to protect personal data, Morocco has aligned itself with leading countries in this area, adopting a legal framework that meets international standards. Additionally, the country has established control and oversight bodies to ensure the protection of personal data. Law 09.08, in particular, mandates that all data processors adhere to key principles such as obtaining prior consent, ensuring the right to access and information, and safeguarding the right to rectification and exposure.

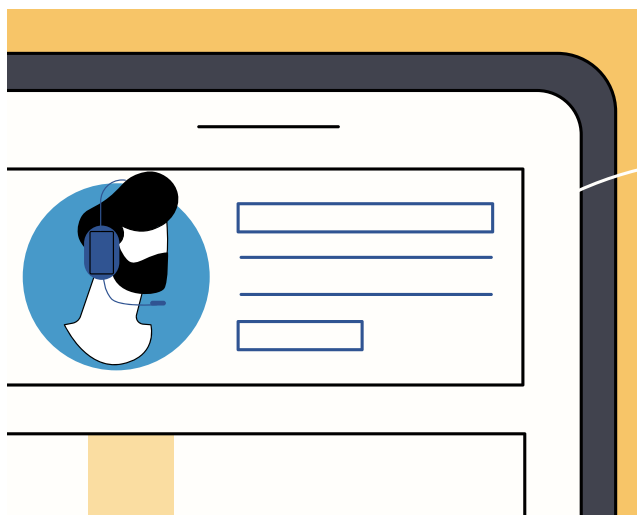
²⁰ Law 53-05 on the electronic exchange of legal data.

<https://www.cndp.ma/images/lois/loi-53-05-echange-electronique-donnees-juridiques-ar.pdf>

²¹ Law 53-05 on the electronic exchange of legal data. <https://www.cndp.ma/images/lois/loi-53-05-echange-electronique-donnees-juridiques-ar.pdf>

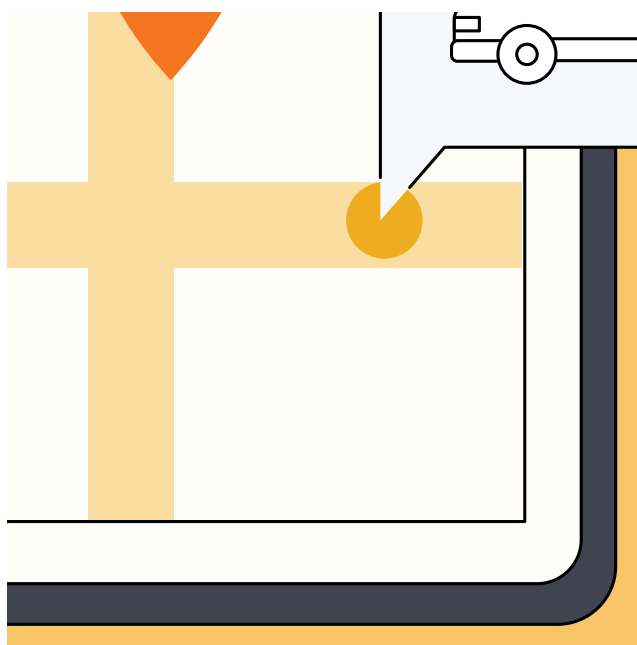
²² Law 03.07 on the disruption of the functioning of automated data processing systems. <https://www.dgssi.gov.ma/fr/loi-07-03-completant-le-code-penal-en-ce-qui-concerne-les-infractions-relatives-aux>

²³ Press Law, Articles 38, 44 to 51 bis, <https://adala.justice.gov.ma/reference/html/Ar/65220.htm>



III. DIGITAL HEALTH TRANSFORMATION: NEW STRATEGIES AND POLICY ACTIONS

Morocco has implemented several strategies to elevate its digital sector, such as the Morocco Digital 2020 Plan, which aims to position the country among the top three digital leaders in the MENA region by accelerating digital transformation, including in healthcare. A key part of this initiative was the creation of the Digital Development Agency in September 2017. The government also adopted a national plan for administrative reform (PNRA) 2018-2021, which focuses on developing new technological tools and expanding them across administrative services to enhance common IT applications and thus streamline access.



The development of a Master Plan for the Digital Transformation of Public Administration is one of Morocco's key initiatives, alongside the establishment of the "Government Gateway" platform to facilitate access to data collected from administrative services, and thereby, access to services. Another important pillar is the Citizen Complaints and Grievances Management System, which strengthens communication channels between users and the administration, including in the health sector.²⁴

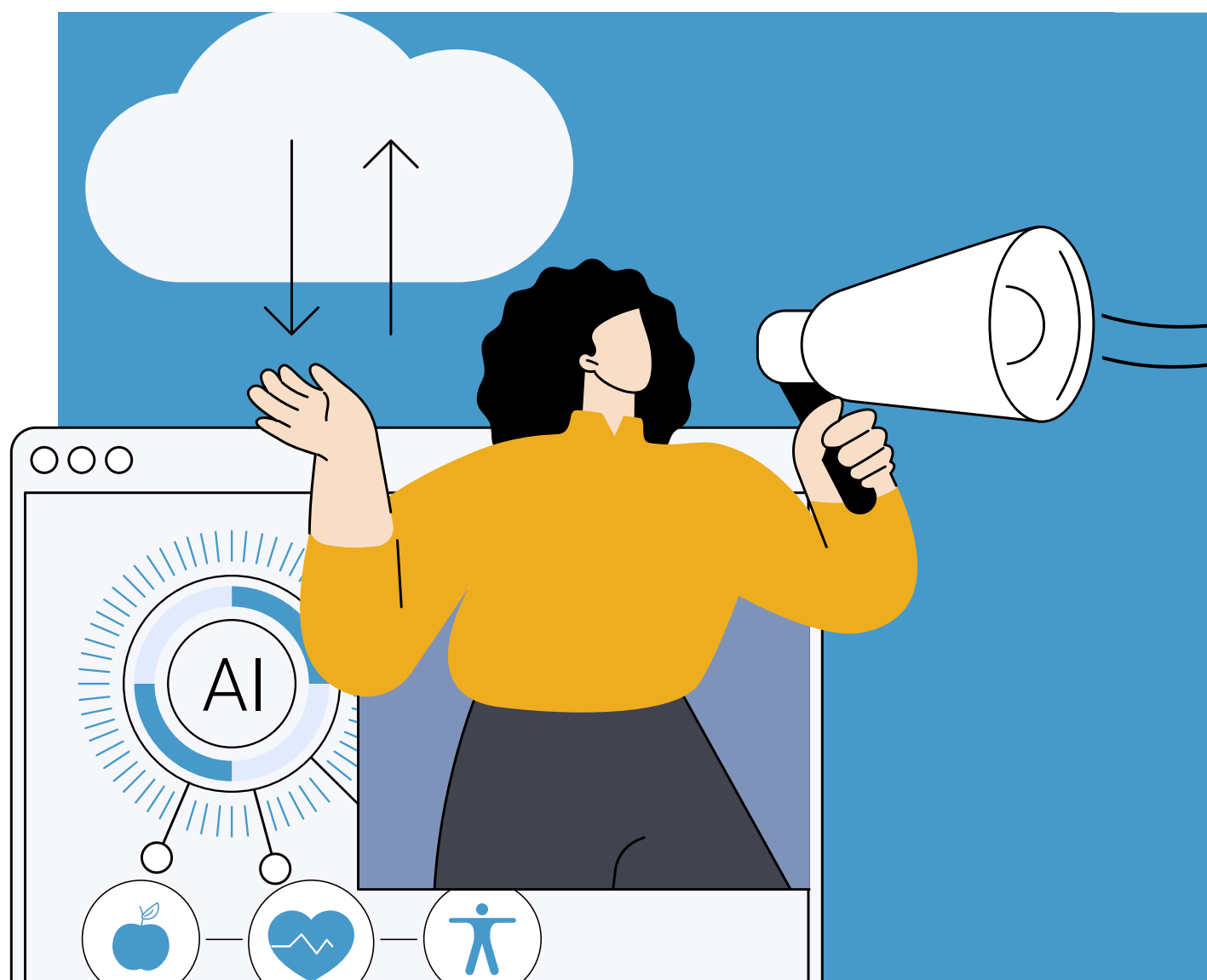
The government also created the Agency for the Digital Development in Morocco, which operates under the ministry delegate to the head of government in charge of digital transformation and administrative reform. The agency is responsible for supporting and implementing the state's digital strategy and Industry 4.0 revolution and promoting digital tools and their use among citizens.²⁵ Its responsibilities include structuring the digital ecosystem, creating real operators in the digital economy, contributing to the promotion of digital administration by guaranteeing harmonious communication channels between citizens and businesses, and establishing a normative framework for digital products and services.

The agency focuses on the digitalization of public and private health organizations and on the use of AI for health data governance.

Morocco has also implemented mechanisms like the "Cadre Général d'interopérabilité" and the National Committee for the Simplification of Administrative and Procedural Procedures. In 2022, Morocco launched two projects aimed at defining a unified strategy for the governance of public data. The first is the National Population Register (RNP), which ensures the reliability of identification data and authentication services for beneficiaries of social programs, benefitting the second project, the Single Social Register (RSU). The RSU is designed to then register applicants for the various social programs based on the data it receives from the RNP.

²⁴ Ministère de la Réforme de l'Administration et de la Fonction publique, *Plan National de la Réforme de l'Administration 2018-2021*, 41, <https://hazbane.asso-web.com/uploaded/plan-national-de-la-ra-forme-de-la-administration-2018-2021.pdf>

²⁵ Agence de Développement Digital, *Présentation de l'ADD* (2017), <https://www.add.gov.ma/presentation-de-ladd>



IV. AI AND HEALTH DATA GOVERNANCE IN MOROCCO: THE CASE STUDY

According to the World Health Organization (WHO), Morocco made significant improvements in recent years. The Health Ministry launched a national e-health system aimed at providing citizens with equal access to medical records and other online services. This e-health system is central to the ongoing debate on health data governance in Morocco. This session focuses on the five main data governance practices — data collection, processing, sharing, storage, and security — organized around the three main objectives of the theoretical framework — protect people, promote the value of health, and prioritize equity — and their supporting pillars.

Data Collection

P1. Protect People

AI's role in data governance has grown significantly, particularly since COVID-19, further complicating an already complex field, especially regarding the very definition of "data." Health data, encompassing both personal and clinical information, exemplifies this complexity, as a primary challenge arises from the absence of a universally accepted definition of data. The World Bank Group highlights the multifaceted nature of data: It "may be quantitative or qualitative; stored on analog or digital media; collected through observation, as part of digital transactions or simply as by-products of digital lives; processed, structured, and analyzed to be converted into information (data are not



synonymous with information); about people, things, and systems.”²⁶

P.1.1 Protect Individuals & Communities

In the context of protecting individuals and communities in health data collection, Law No. 09.08 establishes essential safeguards for personal data processing. It ensures that individuals' health information is handled responsibly and with respect for privacy.

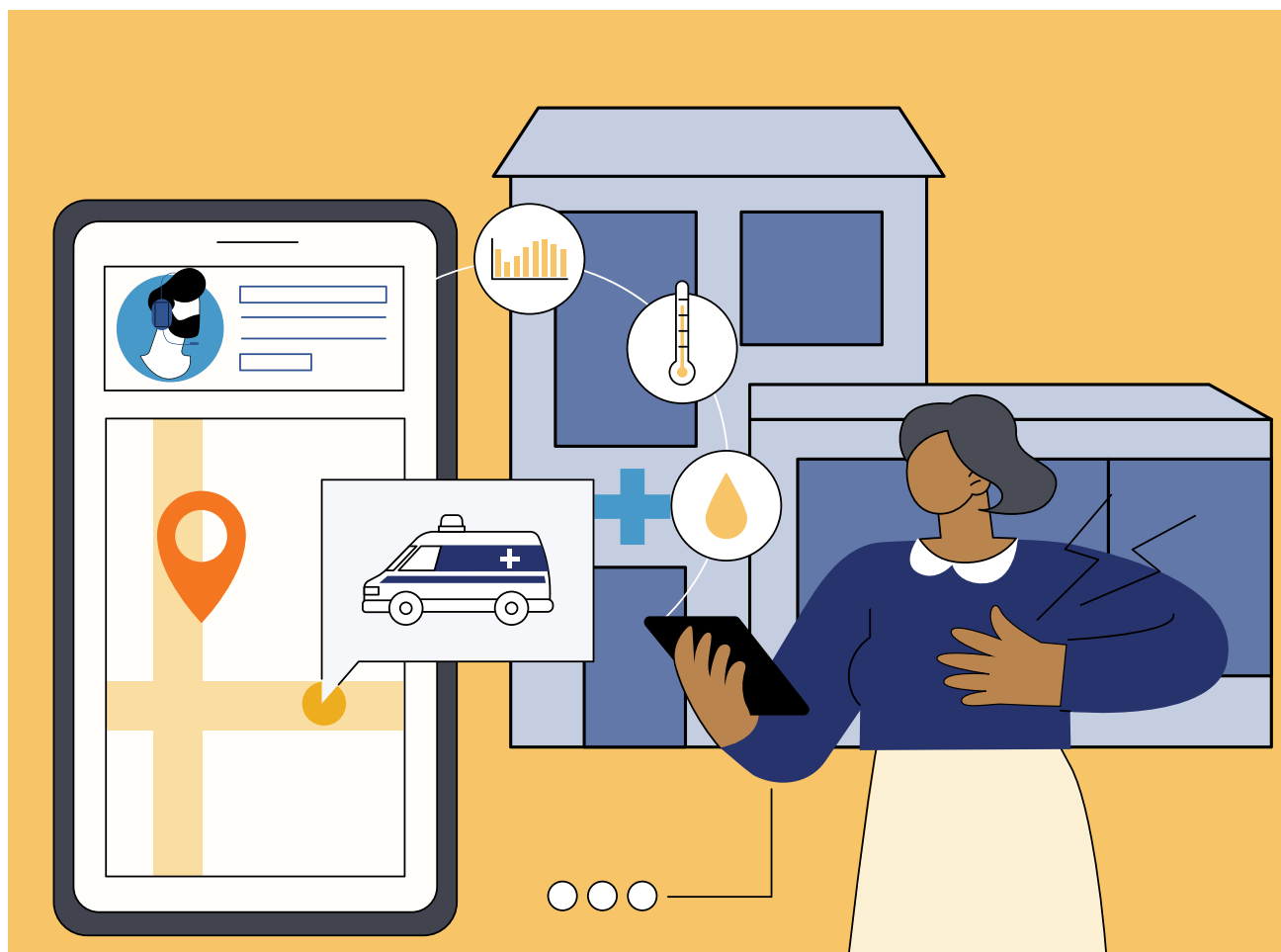
Specifically, in Section 2, titled “Data Quality and Prior Consent of the Data Subject,” Article 3, Paragraph 1-b, the law states that personal data must be collected for specific, explicit, and legitimate purposes and must not be processed later in a way that is incompatible with those purposes. This provision is critical in ensuring that health data is used ethically and within

each patient, in both paper and digital formats, are maintained, with a unique assigned to each patient. This number provides access to both the paper and digital records. Digital tools used to protect data are not connected to the internet for general staff access and are only accessible to authorized personnel. The combination of paper-based and digital methods is essential, particularly during the digital transformation process, to ensure a high level of data storage and protection in line with the legal framework. Subsequently, collected data is secured offline for general staff, with access restricted solely to employees with proper authorization.

P.1.3 Ensure Data Security

The protection of personal data is regarded as a modern and active right, with programs established to protect individuals during the processing of their personal data.

²⁶ World Bank, *Implementation Knowhow Brief* (2021), 9, <https://openknowledge.worldbank.org/server/api/core/bitstreams/2dad9719-f398-4414-ae58-e4f3eca16f26/content>



Digital applications and a specialized laboratory program connected to an internal server are used for data collection in government initiatives such as healthcare insurance programs (RAMID, CNOPS). Such applications and programs such as HOSIX are also used to collect data collected from patients not covered by these initiatives.

Electronic databases are interconnected, ensuring uniformity of data. Special attention is given to sensitive data, such as cases involving patients with infectious diseases like AIDS, to ensure enhanced protection and confidentiality.

P.2. Promote Health Value

Several digital and manual measures are adopted to ensure high-quality data collection through data verification, such as confirming that weight measurements are recorded in the correct field instead of height, or ensuring the data belongs to the same patient. These measures include:

- Reviewing the data;
- Ensuring accuracy in diagnosis;
- Restricting digital program progression unless required fields, such as file numbers and national ID numbers, are completed.

To further ensure accuracy, health institutions utilize digital tools, with data reviewed once by the responsible party.

Several problems and challenges were identified by all interviewees during the data collection process, whether using paper-based or digital methods. Issues include patients lacking a national ID card, leading to imprecise age records, and surnames being inaccurately written. These challenges persist despite the quality of the program used and its primary goal of providing high-quality service.

P.2.1 Enhance Health Systems Services

Morocco has implemented several strategies to elevate its digital sector, such as the Morocco Digital 2020 Plan, which aims to position the country among the top three digital leaders in the MENA region by accelerating digital transformation, including in healthcare. A key part of this initiative was the creation of the Digital Development Agency in September 2017. The government also adopted a national plan for administrative reform, “Plan National de la réforme de l’administration (PNRA),” 2018-2021, which focuses on developing new technological tools and expanding them across administrative ser-

vices to enhance common IT applications and thus streamline access.

The development of a Master Plan for the Digital Transformation of Public Administration is one of Morocco's key initiatives, alongside the establishment of the "Government Gateway" platform to facilitate access to data collected from administrative services to facilitate access to services. Another important pillar is the Citizen Complaints and Management System, which strengthens communication channels between users and the administration, including in the health sector.²⁹

P.2.2 Promote Data Sharing & Interoperability

Morocco has also implemented some mechanisms such as "Cadre Général d'interopérabilité" and the National Commission for administrative simplification. In 2022, Morocco launched two projects to define a unified strategy for the governance of public data. The first is the National Population Register (RNP), which ensures the reliability of identification data and authentication services for beneficiaries of social programs, benefitting the second project, the Single Social Register (RSU). The RSU is designed to then register applicants for the various social programs based on data it receives from the RNP.

P.2.3 Facilitate Innovation Using Health Data

Morocco is fostering innovation using health data through the deployment of laboratory information systems for medical dispatching and biological sample collection.

During the COVID-19 pandemic, the government implemented a smart vaccination management system that utilizes QR codes and SMS notifications to track and monitor the immunization campaign. Additionally, the government promoted telemedicine to facilitate access to health services via remote consultations, diagnosis, and follow-up for suspected and chronic patients, which is exemplified by the platform DABADOC.

In the same period, Morocco started investing in artificial intelligence and leveraging big data to analyze viral genomic surveillance and epidemiological trends. In a sense, the government encourages partnership with multiple stakeholders, particularly start-ups and civil society, to develop digital solutions for health data collection.

P.3. Prioritize Equity

Prioritizing equity in health data collection in Morocco is crucial for achieving universal health coverage and reducing health inequali-

ties. According to the World Health Organization, health equity ensures that everyone has a fair opportunity to reach their full health potential, without being disadvantaged by their social position or other socially determined circumstances.²⁷ Health data collection plays a vital role in monitoring health equity and informing policies and interventions that address the social determinants of health.²⁸ The adoption of health information systems offers a practical solution for improving health equity, as it enables data collection from diverse sources such as laboratories, insurance schemes, civil society and research institutions.

By emphasizing the necessity of equity in health data collection, Morocco can more effectively identify and address the health needs of its population, particularly among disadvantaged groups excluded from accessing quality healthcare services.

P.3.1 Promote Equitable Benefits from Health Data

E-health is an area where AI can have a positive impact, as it leverages digital technologies to enhance health care and services. Since COVID-19, several initiatives have been implemented, including the Wiqaytna app, which enables authorities to trace potential COVID cases.²⁹

To address the fragmented e-health system currently managed by university hospitals, regional health authorities, and various national health programs, a unified system is being developed.³⁰ This system will comply with the country's legal framework for personal data protection and the regulation of medical practice.³¹

P.3.2 Establish Data Rights & Ownerships

The participants in this case study include actors from both public and private sector organizations. These include CHU Mohammed VI Hospital, urban and rural dispensaries, private clinics, Hospital Mohammed VI Tahanaout, Sidi Abdellah Ghiat Health Center, a center for the

27 Kawtar Zahidi et al, "Universal Health Coverage in Morocco: The Way to Reduce Inequalities: A Cross-sectional Study," *The Open Public Health Journal* 15 (2022), <https://openpublichealthjournal.com/VOLUME/15/ELOCATOR/e187494452212220/FULLTEXT/>

28 Tarek Drissi Bouzaidi, Aziz Ragbi, "An Analysis of the Trend towards Universal Health Coverage and Access to Healthcare in Morocco," *Health Economics Review* 14, no. 5 (2024), <https://doi.org/10.1186/s13561-023-00477-0>

29 OECD.AI, *Strategy for Developing an AI Ecosystem* (2021), <https://oecd.ai/en/dashboards/policy-initiatives/http:%2F%2Faiipo.oecd.org%2F2021-data-policyInitiatives-27136>

30 Sofia Alypova, "E-Government Development in Morocco" (2023), *Center for African Studies, HSE University*, <https://we.hse.ru/en/irs/cas/passma>

31 Ruben Tchounyabe, "Morocco Projects National E-Health Integrated System," (2023), *We Are Tech Africa*, <https://www.wearetech.africa/en/fils-uk/news/public-management/morocco-projects-national-e-health-integrated-system>

TYPE OF DATA/ CLASSIFICATION	THE CONTENT	TOOLS
Personal health data	Surname/first name, date of birth, expiry date, identification number/ social security number, and biometric data, such as finger prints	Digital and analog
Health records: Medical/clinical data	Medical history, disease progression, family environment, body temperature, weight, blood pressure, diseases, existing illnesses, prescribed medications, genetic and rare diseases, and insurance (such as Ramed or CNOPS)	Digital and analog
Demographic data related to health	Age, Economic Status, Race and Gender	Digital and analog

Table 1: Health Data Collection

diagnosis and treatment of respiratory diseases, as well as laboratories, pharmacies, physicians, and patients.

Participants from these organizations confirmed that the system facilitates data collection from the initial consultation or diagnosis through examination and follow-up, particularly in terms of electronically saving data and maintaining patients' medical records.

Participants to this study provided evidence regarding data collected, such as their names, addresses, phone numbers, and family environment, as well as demographic and health record data, including information about diseases, any illnesses they suffer from and subsequently prescribed medications, genetic and rare diseases, medical history, disease progression, past illnesses, symptoms, temperature, and more (Table 1).

Health data is collected for specific purposes such as communication with and tracing patients, identifying target groups, and comparing them to the patients' conditions. Regulations regarding health data protection and privacy are well understood by all frontline health

workers involved in data collection. Each analysis is associated with data including age and gender. According to data managers, 18.8 percent reported that data is collected using paper, 12.5 percent confirmed that digital tools such as hard drives and apps are used, and 68.8 percent indicated that both paper and digital tools are used (Figure 2).

In rural regions, 74.4 percent of data managers said that data is collected using paper, compared to 25.6 percent in urban regions. In contrast, 79.7 percent of data managers in urban areas use digital tools for data collection, compared to only 20.3 percent in rural regions.

The digital program used in health organizations ensures a high level of data accuracy. In the case of errors, such as incorrect dates or national ID numbers, the program issues a warning to minimize the margin of error. When an error is detected, it is corrected by retrieving the patient's data, reviewing it, and confirming it in the application where the patient is registered.

Several digital and manual measures are implemented to ensure high-quality data collection through verification processes (e.g., ensuring

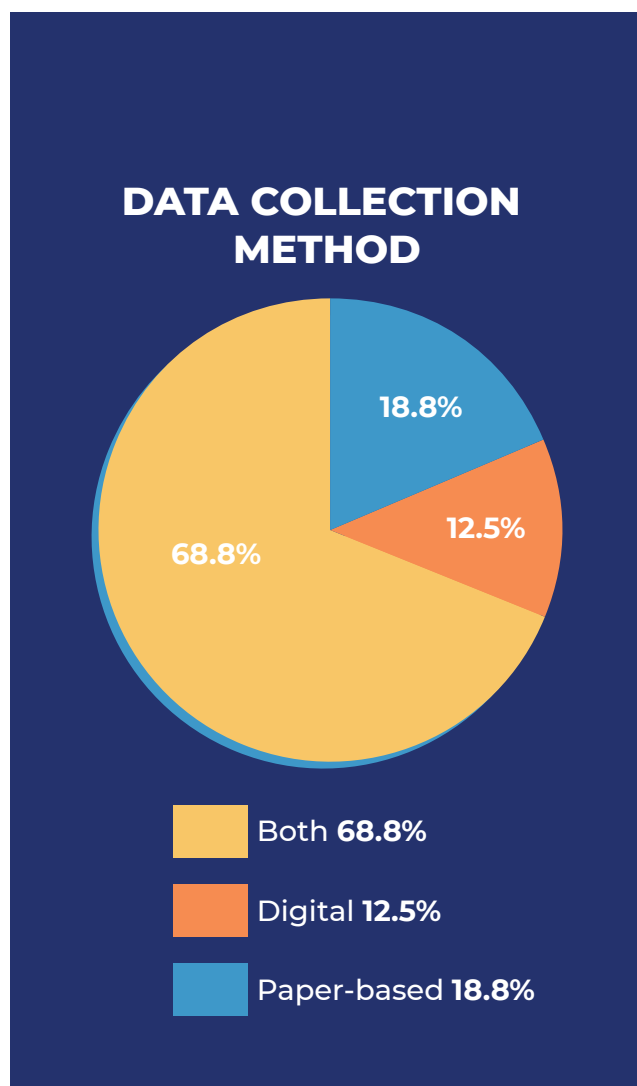


Figure 2: Data Collection Methods

weight measurements are recorded in the designated field and not height and vice versa, or confirming that the data belongs to the correct patient). These measures include reviewing the data and ensuring diagnostic accuracy. The digital program allows editing only until the required fields (file number and national ID number) are completed. To ensure accuracy, health institutions utilize digital programs and the data is reviewed once by the responsible party.

A number of challenges were identified by all interviewees during the data collection process, whether using paper-based or digital methods. For instance, when a patient lacks a national ID card, it results in the documentation of an inaccurate age and potentially incorrect surname spelling. These issues persist despite the adoption of quality programs.

Data processing

P1. Protect People

Law No. 09.08 regulates the protection of individuals concerning the processing of personal data, including health-related data. In the same context, Decree 1026-97-2, issued on 27 Shawwal 1418 (February 25, 1998), establishes the general conditions for the operation of public telecommunications networks, as referenced in the previous section.³² Additionally, the Penal Code addresses offenses related to automated data processing systems, specifically in Articles 442 to 445, as well as Article 448 of the Criminal Code.³³

P.1.3 Ensure Data Security

The protection of personal data is seen as a modern and active right, with programs established to protect individuals during the processing of their personal data. Data processing must comply with the basic components of personal data protection, namely independent supervision and respect for the rights of the data subject.

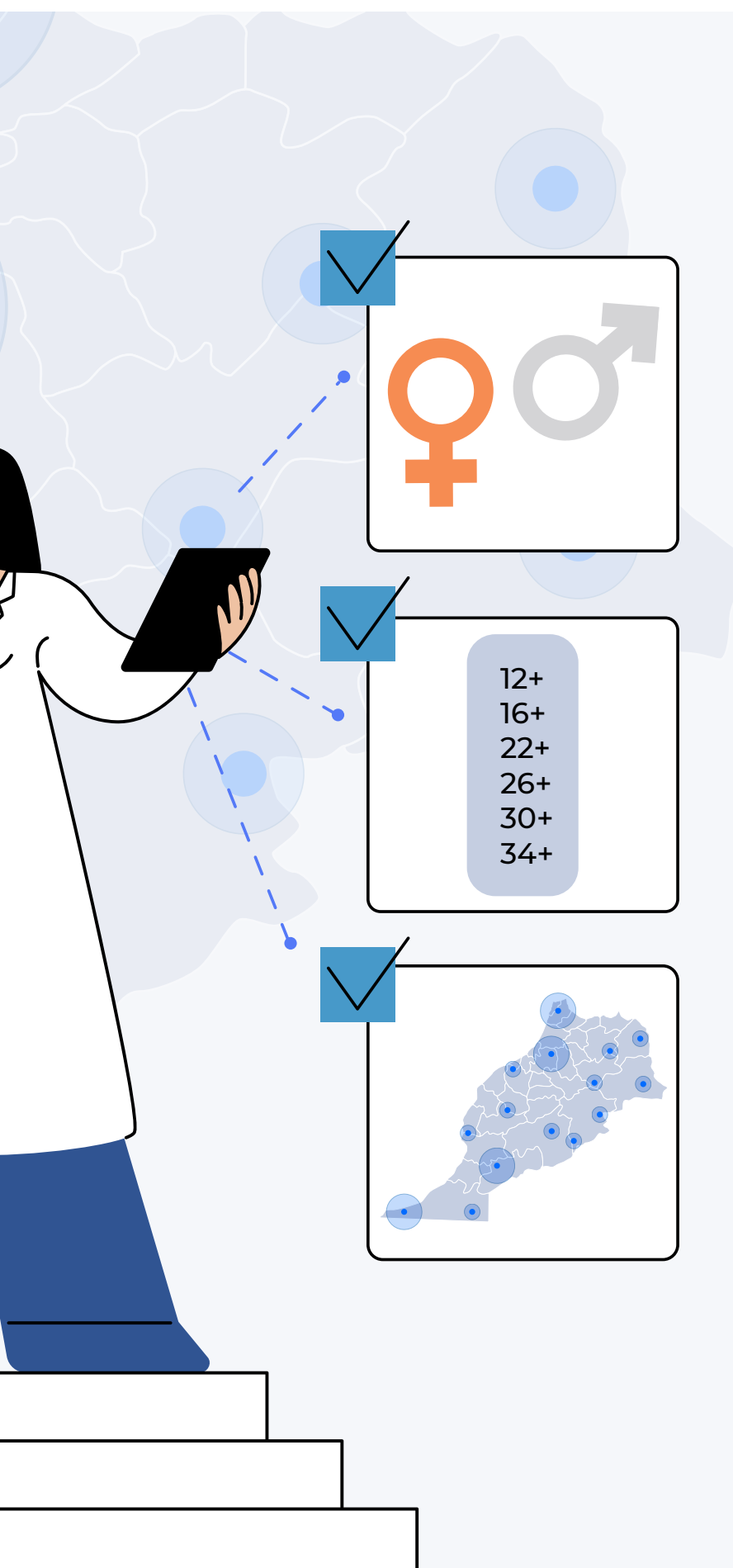
According to representatives from targeted health institutions, artificial intelligence and emerging technologies could significantly aid in data processing. These technologies are seen as a promising solution for improving data management, and interviewees are open to exploring future AI innovations that could simplify data handling and enhance efficiency and accuracy. Interviewees believe AI could help address current challenges related to data accuracy and bolster the skill of human resources, particularly through training and the transition from traditional digital systems to AI-based solutions. Raising public awareness about AI is also essential to overcoming data management challenges. To address these challenges, the best approach is based on four pillars:

- Updating current technological programs to incorporate new AI innovations.
- Conducting awareness campaigns for citizens.
- Providing training for human resources, especially data managers.

³² Decree2-97- 1025 of 27 Shawwal 1418 (February 25, 1998), relating to the General Conditions for the Exploitation of Public Telecommunications Networks, as amended by Decree No. 2-05-771 of 6 Jumada II 1426 (July 13, 2005), and Decree No. 2-06-418 of Ramadan 17, 1427 (October 10, 2006), <https://www.anrt.ma/en/decree-ndeg-2-97-1025-27-chaoual-1418-february-25th-1998-interconnection-telecommunications>

³³ Criminal Code, Chapters 442 to 445, Law 03.07 on the Disruption of the Functioning of Automated Data Processing Systems, accessed August 21, 2024. <https://www.dgssi.gov.ma/sites/default/files/legislative/brochure/2023-03/loi%2007-03.pdf>





- These measures, aimed at improving data processing, are further supported by additional actions to ensure data protection and security.

P.2. Promote Health Value

More than 90 percent of the responsible or heads of health institutions in both public and private sectors said that the above strategies, such as the E-Health System for example, are implemented, and that this is a part of the Moroccan government's effort to support digital transformation across sectors, including in health. An interviewed expert figure in health institutions confirmed the use of both digital and analog documentation methods during all data governance processes.

P.2.3 Facilitate Innovation Using Health Data

The Moroccan government created the Agency for Digital Development under the supervision of the ministry delegate to the head of government in charge of digital transition and administrative reform. The agency is responsible for supporting and implementing the state's digital strategy and Industry 4.0 revolution and promoting digital tools and their use among citizens.³⁰

The agency focuses on the digitalization of both public and private health organizations both public and private sector, with an emphasis on using AI for health data governance. It is also responsible for contributing to the promotion of digital administration by guaranteeing harmonious communication channels between citizens and businesses), and establishing a normative framework for digital products and services.

P.3. Prioritize Equity

Digital tools are utilized in both public and private sectors, as well as in urban and rural regions, as confirmed by 97.2 percent of the associations participating in this case study.

Interviews with responsible parties confirmed that health organizations use digital tools such as the HOSIX program for patient data collection, archiving, and case tracking.

These tools help improve patient access to health services by streamlining appointment scheduling for each patient.

P.3.1 Promote Equitable Benefits from Health Data

Health institutions are investing more in enhancing their performance to address these challenges, focusing on strengthening human resources and collaborating with relevant authorities to further advance the digitization process. According to 92.9 percent of those interviewed — directors, data managers, doctors in the private sector and in public associations, and patients — the use of artificial intelligence in data management can help overcome these challenges, particularly when employed by laboratory technicians during the pre-analytical phase. Doctors interviewed stated that artificial intelligence is a future-oriented project that many in the field aspire to integrate into their practice.

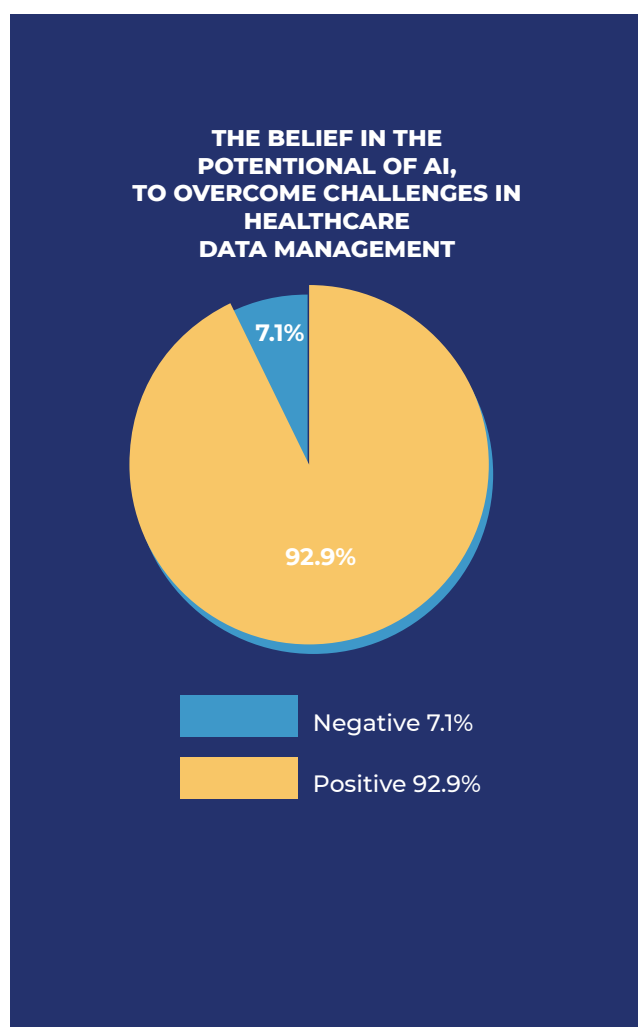


Figure 3: Responses on perspectives on the potential of AI to overcome challenges in health data management

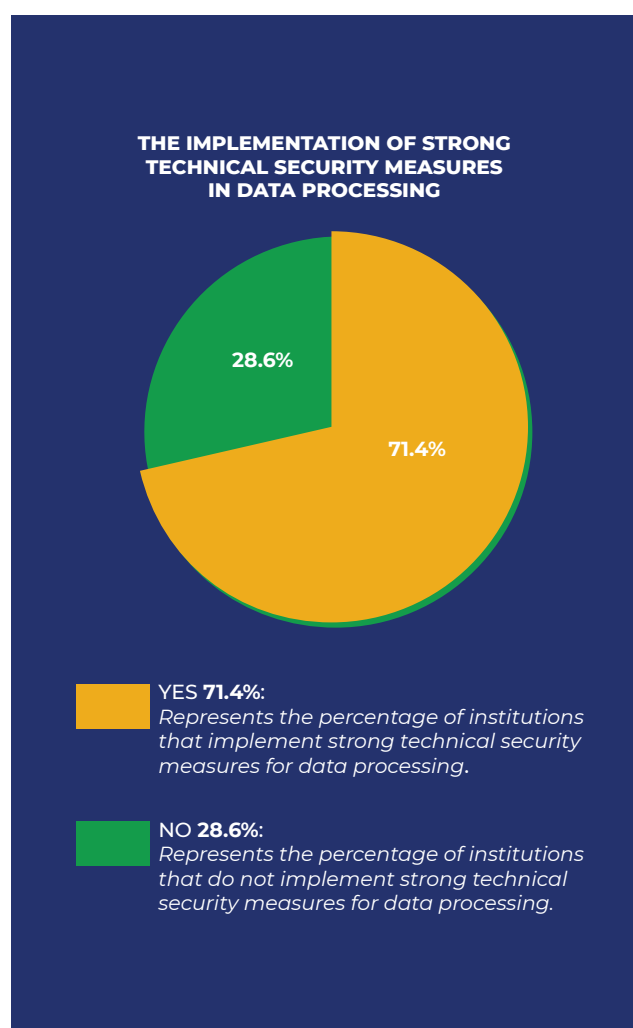


Figure 4: Responses on the implementation of strong technical security measures for data processing

Digital technologies are used in all health institutions for data processing, especially HOSIX, which plays a critical role in assisting data managers at various levels, particularly by reducing effort and saving time. This tool aids in management and enhances the quality of services provided. To ensure secure data processing, 71.4 percent of institutions — including public and private hospitals in rural and urban regions, associations, and labs — have implemented strong measures, such as requiring passwords, obtaining authorizations from institutional heads, and tracing all activities related to data processing (Figure 4).

The remaining 28.6 percent of institutions are generally small public medical centers in rural and urban regions that use paper or more traditional digital tools, and need to update their tools and reinforce their methods.

P.3.2 Establish Data Rights & Ownerships

According to article 4 of law 09-08, “The processing of personal data can only be carried out if the person concerned has undoubtedly given their consent to the operation or to all of the operations envisaged.”

The interviewer confirms that the consent is mandatory for data collection, processing, and storage. However, Moroccan legislators have made exceptions to this rule. Consent is not required in specific cases, such as when data processing complies with another legal provision or when it is necessary for the execution of a mission of public interest or related to the exercise of public authority.

Data Sharing

P1. Protect People

The legal provisions in the first paragraph of Article 5 address the conditions of confidentiality and neutrality regarding transmitted messages and the protection of beneficiaries’ private lives and personal data.³⁴ This rule can only be violated under conditions explicitly established by law.

Operators are obliged to inform their employees about their obligations and the penalties for

breaching confidentiality, including in matters related to health correspondence.

Law 103.13 on combating violence against women³⁵ introduces amendments to the Moroccan Criminal Code to address privacy violations, especially concerning the misuse of information systems or other means for nonconsensual capture, recording, broadcasting, or distribution of statements or information intended to remain private or confidential. The penalties for such acts include imprisonment ranging from six months to three years and a fine between 2,000 and 20,000 dirhams. These penalties also apply to individuals who intentionally capture, record, broadcast, or distribute images of a person in a private setting without his consent.³⁶

P.1.2 Build Trust in Data Systems

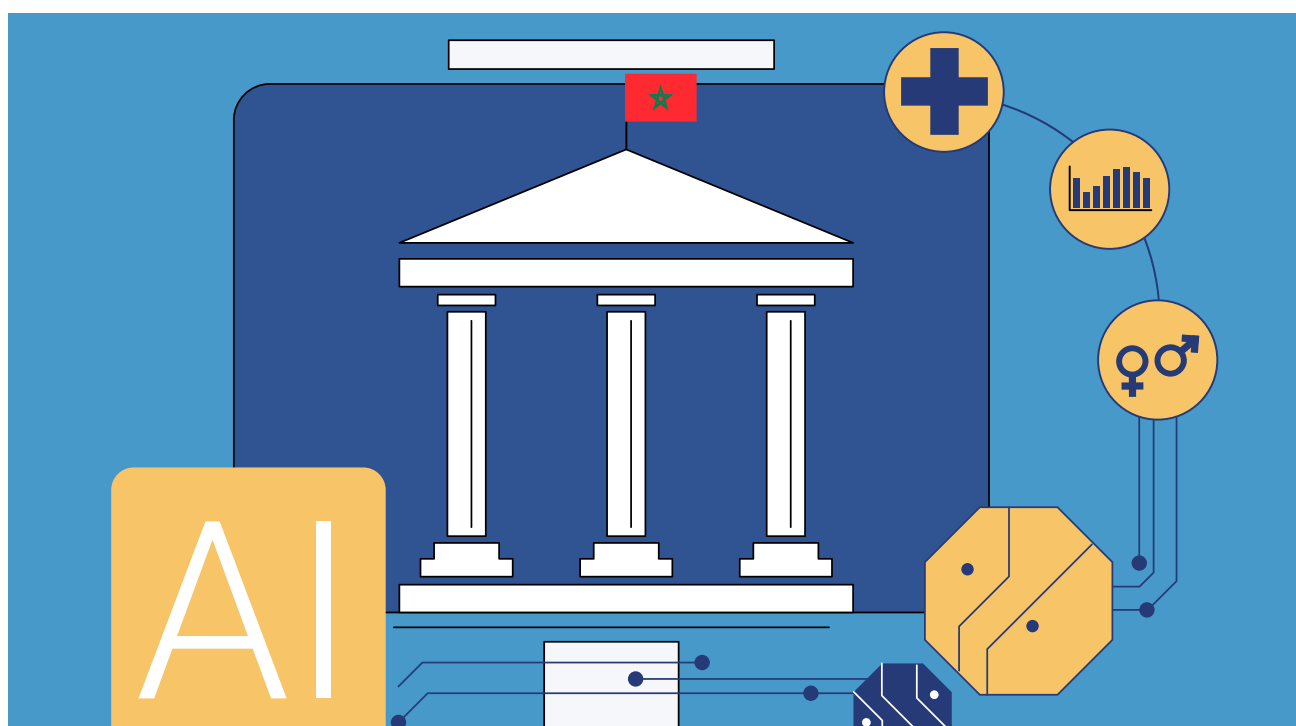
The framework for data sharing within the healthcare system in Morocco is regulated by law, with procedures designed to ensure secure and appropriate sharing of health data.. Virtual Private Networks (VPNs), antivirus software, and anti-spyware systems are also implemented to enhance security.

In some cases, health data is primarily shared with treating physicians to facilitate diagnosis and treatment. In cases involving minors, chil-

34 Decree 2-97-1025 of 27 Shawwal 1418 (February 25, 1998), relating to the General Conditions for the Exploitation of Public Telecommunications Networks, as amended by Decree No. 2-05-771 of 6 Jumada II 1426 (July 13, 2005), and Decree No. 2-06-418 of Ramadan 17, 1427 (October 10, 2006). <https://www.anrt.ma/en/decrees-decree-2-97-1025-27-shawwal-1418-february-25th-1998-interconnection-telecommunications>

35 Law 103.13 on combating violence against women, with special provisions for the protection of privacy added to the Moroccan Penal Code, <https://social.gov.ma/%D8%A7%D9%84%D9%85%D8%B1%D8%A3%D8%A9/%D8%A7%D9%84%D9%82%D8%A7%D9%86%D9%88%D9%86-103-13-%D9%88%D9%85%D8%B1%D8%B3%D9%88%D9%85%D9%87-%D8%A7%D9%84%D8%AA%D8%B7%D8%A8%D9%8A%D9%82%D9%8A/>

36 Ibid.



dren, or and persons with mental disabilities, data may also be shared with family members. If necessary, additional health information can be requested from healthcare institutions, including laboratory results, radiology reports, or data directly from the patient.

P.1.3 Ensure Data Security

Data collection for government initiatives such as healthcare insurance programs (RAMID, CNOPS) utilizes specialized digital applications and laboratory programs connected to internal servers. This ensures the data is not exposed to the internet to maintain high levels of security. Access to these systems is restricted to authorized personnel only. HOSIX is also employed for collecting and managing data not associated with these insurance programs.

P.2. Promote Health Value

Morocco promotes the health value through the following initiatives:

- The development of a master plan for the digital transformation of public administration
- The establishment of the Government Gateway platform to allow common access to data from administrative services and facilitate access to services.
- The development of a Citizen Complaints and Grievances Management System to strengthen communication channels between users and the administration in all sectors, including health.²⁹

P.2.1 Enhance Health Systems Services

Collecting information through the national health information system and transferring it to the digital system facilitates easier study and analysis, as well as coordination between local and national health institutions.

Data is reviewed based on need, whether for invoices, appointments, or use by medical students (in the context of research and statistics). This review is conducted once or multiple times by the person entering the information into a software system such as HOSIX, while adhering to established rules and administrative frameworks.

P.2.2 Promote Data Sharing & Interoperability

Promoting data sharing and interoperability is ensured through coordination with various stakeholders, such as the Health Ministry, police, and courts (for example in cases involving women victims of violence). This coordination neces-

sitates monthly data updates. Additionally, training for professionals and the reinforcement of digital transformation are achieved through the use of mobile applications for communication.

P.3. Prioritize Equity

P.3.1 Promote Equitable Benefits of AI

All the strategies related to health data governance are grounded in a gender-sensitive approach. The legal framework emphasizes equity, designed to protect all individuals. The implementation of these laws further emphasizes equity, aiming to provide equal benefits for all. For instance, government initiatives, managed in collaboration with civil society and NGOs, address sexual and reproductive health issues and gender-based violence, specifically among women migrants in both urban and rural areas.

Some of the strategies that Morocco has adopted to prioritize equity in health data collection include:

- Training for health workers on gender, equality, and data collection, focusing on disaggregating data by sex, age, geographic location, and other dimensions of difference.
- Promoting the dissemination and communication of health data, coupled with training health workers on the use of AI tools.
- Integrating all stakeholders, including health professionals, policymakers, and civil society, in raising awareness and advocating for health equity.
- Establishing connections between actors in health data, women's associations, and policymakers to foster collaboration and better coordination.

P.3.2 Establish Data Rights & Ownerships

The government seeks to strike a balance between privacy and data protection laws and the right to access information. Law 09-08, specifically Article 1, paragraph 9, defines "consent of the data subject" as a free, specific, and informed expression of will by which an individual agrees to have their personal data processed. Article 11 of the law addresses necessary conditions for the "interconnection of data," which involves linking data from one file to another, whether held by different entities or by the same entity for a different purpose. This law applies to both automated and non-automated processing of personal data, including manual files, as outlined in Article 2, paragraph 1.



Data Storage

P1. Protect People

Data is stored using the digital system and copies are made on a hard disk, USB drive, or servers based in Morocco or abroad.

P.1.1 Protect Individuals & Communities

The archive retention period for paper-based data is two years, whereas digital data is stored in a closed archive secured with a key, accessible to only three authorized individuals. To ensure confidentiality and security, institutions utilize encryption, and physical access is restricted by a locked room. Data is stored for one year under these conditions.

Data protection efforts begin at the storage level, with institutions adopting various to safeguard both paper and digital records. In this context, both directors and data managers follow the establishment of data sharing frameworks (such as infrastructure for data systems) across health-care systems to support data sharing.

P.1.2 Build Trust in Data Systems

The combination of paper-based and digital methods is necessary, particularly during the digital transformation process. This combination ensures proper data storage and protection while complying with the relevant legal framework.

P.1.3 Ensure Data Security

Data security and the protection of patient privacy is ensured also through de-identified and/or encrypted data, with access restricted to only the treating physician and the patient. For security standards, only three authorized individuals are permitted access to the data and institutions use encryption and secure rooms locked with keys.

In addition to the previous rules and standards, data security is ensured through complete confidentiality and extreme caution, including the use of secret numbers and private accounts for security programs. Data security standards and tools are reviewed and audited by the IT department in coordination with the head of the institution.

The safety of valuable information against electronic attacks, fires, and natural disasters is guaranteed through electronic security pro-

grams. Data is stored in remote locations away from risks like fire and natural disasters, with real-time data transfer to servers located outside Morocco every second.

P.2. Promote Health Value

AI can facilitate tasks related to scientific research and identifying population health issues by streamlining the collection and analysis of data from various healthcare institutions. AI has the potential to process this information, turning it into easily exploitable numbers. These emerging technologies could also gather and analyze data according to health goals by global organizations such as the World Health Organization. For successful implementation, the Public Health Ministry should adopt a national health strategy that integrates AI and a gender-sensitive approach.

Such AI technologies are intended to enhance services and improve patients' access to services. In this context, the ministry's investment in digital transformation and patient data can provide more long-term strategies for clinics.

P.2.2 Promote Data Sharing & Interoperability

Providing a health organization with a modern health information system helps enhance communication with patients, thereby fostering trust. The Moroccan health information system is structured around the following health care network:³⁷

- **"Basic Health Care Network"**
 - » *Information subsystem*
 - *Maternal and child health*
 - *Family planning*
 - *School and university hygiene*
 - *Information, education and communication*
 - » *Curative care information subsystem*
 - » *Surveillance information subsystem epidemiological*
- **Hospital management information system SIG-Ho.**
- **Mortality and morbidity (cause of death)**
- **Surveys and Studies**³⁸

This system relies on a combination of both digital and paper-based tools for data collection, such as forms, registers, reports, and loops. It integrates various IT equipment, including PCs

and servers, alongside computer applications (software and software packages), and health databases (infrastructure and services). Communication tools, such as telephones, fax, email, and information bulletins at both central and peripheral levels are also used in this system.³⁹

P.3. Prioritize Equity

Digital tools are used across both public and private sectors and in urban and rural regions, as confirmed by 97.2 percent of associations participating in this case study. Interviews with relevant parties confirmed that health organizations use digital tools, such as HOSIX, for patient data collection, archiving and case tracking. These tools enhance patient access to health services by facilitating the organization of appointments for each patient.

Personal and health data are recorded and stored in HOSIX, which includes all necessary information, such as dates, for the institution's reference. This system also plays a crucial role in organizing appointments for patients.

Responsible interviews noted that HOSIX plays a key role in managing patient appointments, ensuring easier access to health services for all. Using patient ID and digital systems, HOSIX facilitates data collection, accelerates research, and enhances data storage. It also supports maintaining patient confidentiality, as confirmed by all participants in this case study.

P.3.2 Establish Data Rights & Ownerships

The law 09-08 recognizes several rights for individuals, with Chapter two dedicated to "rights of the person concerned." These rights include:

- Right to information during data collection (Article 5), subject to limitations outlined in Article 6
- Right to access data for individuals who provide proof of their identity (Article 7)
- Right to rectification of incorrect data (Article 7)
- Right to object (Article 8), allowing individuals to object to the processing of their data for legitimate reasons upon proving their identity.

Data Security

P1. Protect People

Data managers in all organizations involved in the case study, including health associations, emphasized that personal data is con-

³⁷ Zguiouare Abdelaziz, (2007), *Système National d'Information Sanitaire Cas du Maroc* (2007), Europa, https://ec.europa.eu/health/ph_information/implement/nca/docs/ev_20070703_co07_en.pdf

³⁸ Zguiouare Abdelaziz, (2007), *Système National d'Information Sanitaire Cas du Maroc* (2007), Europa, https://ec.europa.eu/health/ph_information/implement/nca/docs/ev_20070703_co07_en.pdf.

³⁹ Ibid.

sidered sensitive. Data is updated during each visit by the health data manager. This position requires a combination of technical, analytical, and organizational skills, as well as an understanding of healthcare practices and data privacy regulations, due to the vital role that health data managers play in managing and maintaining health-related information.

P.1.1 Protect Individuals & Communities

Patient consent is a necessary component in data collection, with oral consent often sufficient in routine procedures, such as medical treatment. However, explicit and written consent is required for more invasive or specific interventions or when health data is being collected for research purposes. Patients must be fully informed about the study's nature, potential risks, benefits, and their intended use of their data. For electronic health record systems, written consent, along with a copy of the patient's ID is mandatory, especially if the data will be used beyond direct patient care.

P.1.2 Build Trust in Data Systems

Patient consent is required for all patients, which may be verbal or written depending on the situation. In specific situations such as medical treatment or public health surveillance, no consent is required.

According to the interviews, 43 percent of respondents stated they require verbal consent, 43 percent ask for written consent, and 14 percent indicated that no consent is needed (Figure 5).

P.1.3 Ensure Data Security

According to the head of the institution interviewed, only authorized staff can access sensitive patient information. These employees are assigned unique identification numbers and a specific login credential based on their roles and positions. The data is stored on the AIP server. Written authorization is required for accessing sensitive data, particularly related to conditions like sexually transmitted diseases. Patient identity is confidential, and researchers are bound by law to maintain complete secrecy. The law regulates data sharing, ensuring that it is only shared with the authorized individuals previously mentioned or, in the case of minors or death, with the patient's family..

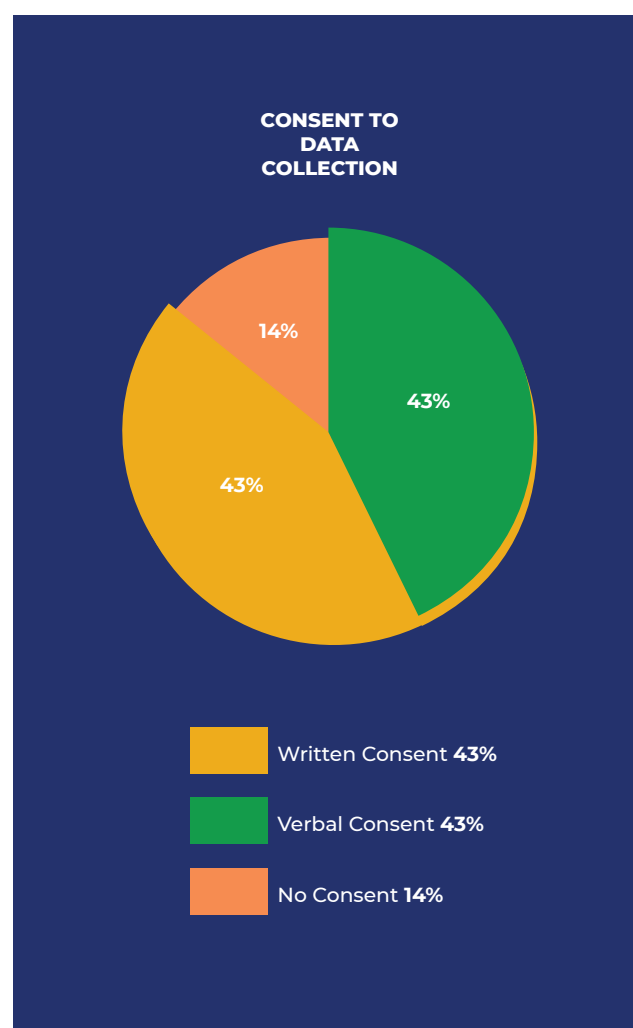


Figure 5: Responses on consent to data collection

ANALYSIS AND TAKEAWAYS: CHALLENGES AND OPPORTUNITIES

Challenges related to health data governance:

According to the heads of the health institutions involved in this case study, several challenges related to health data governance impact access to health services, particularly for women. These challenges include insufficient health infrastructure, such as a lack of internet and electronic devices in some centers. Additional obstacles include issues like incorrect patient data, which may arise from factors such as illiteracy and language barriers, and problems with device connectivity that hinder full coverage in certain areas.

Furthermore, the shortage of human resources and the absence of previously recorded data

make it difficult to rely on paper-based archives, exacerbating the challenges faced in managing and accessing health information.

A significant challenge during the digital transformation period is the reluctance of some citizens and healthcare workers to disclose personal information. The manual, traditional data collection methods in certain centers and the incompatibility between paper-based and electronic data further complicate this issue.

Some patients, particularly women, are hesitant to provide their personal data, fearing a lack of confidentiality. Moreover, the scarcity of equipment, especially in public health institutions, and the lack of continuous training for healthcare professionals, limit citizens' access to healthcare services. These challenges can be attributed to inadequate healthcare funding. In this context, a strategic vision incorporating a gender-sensitive and forward-looking approach is essential to addressing these issues effectively.

CONCLUSION AND POLICY RECOMMENDATIONS

According to the heads of health institutions interviewed in this case study, the most effective strategies to overcome challenges related to health data governance include training, raising awareness, and strengthening data collection systems. There is also a need to support research and innovation and increase funding dedicated to public health in general.

Raising awareness among citizens and incentivizing healthcare workers through targeted awareness programs is crucial for improving health data collection. Additionally, ensuring adequate equipment and a robust digital infrastructure is necessary for successful digital transformation, particularly when combining traditional manual methods with new digital approaches. To support this transition, continuous training for healthcare professionals is essential to ensure they can effectively use digital tools.

Reinvestment in the health sector is crucial to addressing current material resource shortages and adopting policy actions aimed at guiding citizens to trust state institutions. Policy actions are also needed to reinforce the skills of human resources in the healthcare field, and projects as well aimed at providing equal access to devices and the internet in both rural and urban hospitals and medical centers.

Considering trends in AI technologies can offer new opportunities to improve healthcare quality and access and help facilitate scientific re-

search and data collection, processing, storage, safety and security.

The Health Ministry should prioritize investing in such emerging technologies considering that Morocco's health institutions, especially in the public sector, need AI tools to catch up with advances in the field of information and data governance and better provide access to healthcare for citizens and streamline work for employees. Morocco has the opportunity to realize its digital health transformation ambitions by building and reinforcing a robust technical infrastructure, along with a comprehensive policy, legal, and regulatory framework for data governance. This framework will enable trusted data collection, processing, and security, which are essential to a modern, data-driven healthcare system.

However, its successful implementation hinges on strengthening institutional capacity, digital infrastructures, and the skills of all stakeholders within the data ecosystem. To achieve this, an inclusive communication strategy and campaigns is required to increase public awareness and foster trust in state data processing institutions.

As the pace of innovation in emerging technologies accelerates, Morocco must adopt a forward-thinking approach mainly considering the development of a national AI health strategy for 2040. This strategy should ensure health and wellbeing for all, with a specific focus on women in rural areas, as the nation adapts to the era of AI, AGI, and Super Intelligent Systems.



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